MIL-G-46030E
4 June 1984
SUPERSEDING
MIL-G-46030D(MR)
28 September, 1978

MILITARY SPECIFICATION

GLUE, ANIMAL (PROTECTIVE COLLOID)

This specification is approved for use by the Army Materials and Mechanics Research Center, Departmen of the Army, and is available for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 This specification covers one type of animal glue for use as a protective colloid in the manufacture of propellants.

2. APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Specifications, Standards and Handbooks Unless otherwise specified the following specifications, standards and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

RR-S-366	- Sieve, Test.
UU-S-48	- Sacks, Shipping, Paper
MMM-A-100	- Adhesive, Animal Glue

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Director, US Army Materials and Mechanics Research Center, ATTN: DRXMR-SMS, Watertown, MA 02172 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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(Copies of specifications, standards, handbooks, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
D 1193 Reagent Water

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

UNIFORM CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106 South Riverside Plaza, Chicago, IL 60606).

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC.
National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, ATTN: Tariff Order Section, 1616 P Street, N. W., Washington, D. C. 20036).

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies).

2.3 Order of Precedence In the event of a conflict between thetext of this specification and the references cited herein,the text of this specification shall take precedence.

REQUIREMENTS

3.1 First Article. The contractor shall furnish 200 lbs (90 kg) of glue for first article inspection and approval (see 4.5, 6.2 and 6.3). Approval of the first article by the procuring activity shall not relieve the contractor of his obligation to supply glue and containers conforming to this specification. Any change or deviation from the first article sample shall be subject to the approval of the procuring agency.

3.2 MATERIAL. The glue shall be manufactured from raw materials of animal origin

3.3 Physical and Chemical properties.

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3.3.1 Form. The glue shall be supplied in granular or pelletized form, of such size that 95 percent minimum shall pass through a No. 7. U. S. Standard sieve when tested as specified in 4.9.1.2.

3.3.2 Moisture content. The glue shall have a moisture content of not more than 14.0 percent when tested as specified in 4 .9.1.3.

3.3.3 Grease content. The glue shall have a grease content of not more than 6.0 percent when tested as specified in 4.9.1.4.

3.3.4 Ash content. The glue shall have an ash content of not more than 6.0 percent when tested as specified in 4.9.1.5.

3.3.5 pH value. The glue shall have a pH of not less than 5.0 and not more than 7.0 when tested as specified in 4.9.1.6.

3.3.6 Keeping qualities. The glue shall not have an objectionable odor and there shall be no evidence of putrefaction when tested as specified in 4.9.1.7.

3.4 Performance. Samples for the performance test shall be drawn from the preparation of at least 2 hardening batches. The glue shall conform to the following requirements when tested as specified in 4.9.2.

(a) A sample drawn 1 hour after adding the glue to the pilot hardening batch shall show particles of approximately uniform size and shape.

(b) The sample after standing 1 minute shall upon agitation show a redisposition into particles of the same size and shape as those originally shown in the drawn sample.

(c) A sample drawn 2-1/2 hours after the addition of the glue shall meet the same requirements as those of the sample drawn at the end of the first hour.

3.5 Suitability for use with explosives. When suitability for use with a particular explosive is required (see 6.2), a special test shall be conducted at a designated Government laboratory to determine compliance of the adhesive in this respect (see 6.4). The procuring agency shall request this test to be conducted and shall specify the particular explosive to be used in this test.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Lot. A lot shall consist of glue from one batch. A batch is defined as that quantity of material subject to some unit chemical or mixing process intended to make the product substantially uniform.

4.3 Classification of inspection. The inspection and testing of the glue shall be classified as follows:

(a) First article inspection (see 4.5)
(b) Quality conformance inspection (see 4.6)

4.4 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in paragraphs 4.8 and 4.9.

4.5 First article inspection. First article inspection shall consist of inspection for all the requirements of this specification.

4.6 Quality conformance inspection. The contractor shall perform all tests except the performance test (see 4.9.2) The glue may be shipped without Government inspection prior to the performance test at the contractor's risk when authorized by the procuring activity. The contractor shall certify, however, that the glue conforms to all the requirements of this specification except the performance test and that the lot is of the same properties as that of a lot that has previously passed the performance test.

4.7 Sampling.

4.7.1 For examination for preparation for delivery. A random sample of filled sacks of glue shall be selected in accordance with level 1 of MIL-STD-105 to determine compliance with the requirements of Section 5.

4.7.2 Physical and chemical tests. A composite sample consisting of 1 quart (946 ml) of glue representative of the lot shall be selected and placed in a wide-mouth, glass, 1-quart (946-ml) container and sealed tightly and identified as specified in the contract or order.

4.7.3 For performance test. A representative sample consisting of 200 pounds (90kg) of glue shall be selected from each lot for the performance test.

4.8 Examination.

4.8.1 Packaging, packing and marking for shipment. Examination shall be made for the defects and at the acceptable quality level (AQL) as shown in table I to determine compliance with section 5.

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Table I. Methods of examination

Material	AQL Percent	Classification of defect	Defect	Method of Examination
Unfilled sacks	1.0	Major 101	Wrong type (see 5.1)	Visual
Filled sacks	2.5	Major 102	Wrong weight (see 5.1)	Approved scale[1]
		Major 103	Improperly closed (see 5.1)	Visual
		Major 104	Improper Marking (see 5.2)	Visual

[1]Approved by procuring agency.

4.9 Tests

4.9.1 Physical and chemical tests.

4.9.1.1 General. Distilled water in accordance with ASTM D 1193 and reagent-grade chemicals shall be used where applicable throughout the tests. Blank determinations shall be run and corrections shall be applied when necessary.

4.9.1.2 Form. Weigh a representative sample of approximately 100 grams and place on a No. 7, U. S. Standard Sieve conforming to RR-S-366. Pass the sample through the sieve by means of a Ro-Tap shaker or equivalent for 5 minutes, weigh the glue that passes through the sieve, and calculate as follows:

$$\text{Percent through sieve} = \frac{\text{Grams through sieve}}{\text{Grams of sample}} \times 100$$

4.9.1.3 Moisture content. Weigh 10 g of a well-mixed sample in a tared-moisture dish, 60 mm in width and 20 to 30 mm in depth. Heat in an oven at 105°C. ± 1°C (221° ± 1.8°F) for 16 to 18 hours (overnight). After continuous heating, remove from oven, cover lid and place in a desiccator. Cool to room temperature, weigh and calculate:

$$\text{Moisture percent} = \frac{\text{Grams through sieve}}{\text{Grams of sample}} \times 100$$

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4.9.1.4 Grease content. Weigh a 5-gram sample into a 500-ml Erlenmeyer flask with a ground-glass joint. Add 130 ml of water and 20 ml of concentrated hydrochloric acid. Attach a reflux condenser and heat for 4 hours on an electric hot plate whose temperature has been adjusted to about 120° to 140°C (248° to 284°F) to maintain a gentle boil. Cool to room temperature and wash into 600-ml separatory funnel with distilled water. Add about 100 ml of petroleum ether in three portions to the flask and decant into the separatory funnel. Shake the separatory funnel for 2 minutes, releasing the pressure occasionally. Allow the layers to separate, and drain off and discard the aqueous layer. Decant the petroleum ether layer into a tared 150-ml beaker and evaporate to dryness on a steam bath. Dry for 30 minutes at 101° ± 1°C (213.8° ± 1.8°F). Cool to room temperature in a desiccator and weigh. Calculate as follows:

$$\text{Grease percent} = \frac{\text{Grams of residue}}{\text{Grams of sample}} \times 100$$

4.9.1.5 Ash content. Weigh a 1-gram sample into a tared-porcelain crucible. Heat in an oven at 101° ± 1°C (213.8° ± 1.8°F) for 16 to 18 hours (overnight) and then heat in an electric muffle furnace at 550° ± 50°C (1022 ± 90°F) for 16 to 18 hours (overnight). Cool to room temperature in a desiccator and weigh. Calculate as follows:

$$\text{Ash, percent} = \frac{\text{Grams of residue}}{\text{Grams of sample}} \times 100$$

4.9.1.6 pH Value. Weigh into a covered 250-ml. beaker the size sample calculated from the following equation using grams as the unit:

$$\text{Size of sample} = \frac{100}{100 - \text{Percent moisture}}$$

Dissolve this sample in 100 ml. of distilled water by heating at 60°C (140°F). Disregard any white cloudiness that may form. Cool to room temperature and determine the pH electrometrically.

4.9.1.7 Keeping qualities. Weigh a 10-gram sample into a covered 400-mil. beaker. Add 100 ml. of distilled water and heat on a steam bath to about 50°C (122°F) with frequent stirring until the animal glue has dissolved. Disregard the white colloid that appears. Smell the solution. The solution shall be free from any odor of decomposition. Heat the solution in an oven at 40° ± 1°C (104 ± 1.8°F) for 48 hours, and smell the solution again. The solution shall still be free from any odor of decomposition after heating.

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4.9.2 Performance tests. At least 2 hardening batches shall be produced from the samples of glue, in accordance with regular production practice. The amount of glue used in a production-scale batch shall not exceed 60 pounds (27kg). The tests shall be conducted in either pilot-plant scale or production-scale models of hardening equipment normally used in the ball-propellant manufacturing process. One hour after adding the glue to the hardening batch, the sample valve shall be thoroughly flushed and a 1/2-quart (473-ml) to 3/4-quart (709-ml) sample collected in a glass jar. The sample shall then be inspected for size and particle shape. The jar shall be allowed to set for 1 minute. During this period the lacquer particles will rise to the surface of the jar. At the end of this period agitate the jar by swirling, and check for redisposition of particles. Two and one-half hours after the addition of the glue, draw a sample in the same manner as that for the sample drawn after 1 hour. Inspect the sample for size and particle shape.

4.10 Rejection.

4.10.1 Examination. Any nonconforming sample unit when examined in accordance with 4.8 shall be rejected, and if the nonconforming units exceed the acceptance number for that sample, the entire lot shall be rejected.

4.10.2 Tests. A lot shall be rejected for failure to comply with any of the test requirements when tested in accordance with 4.9.

5. PREPARATION FOR DELIVERY

5.1 Packing Packing shall be level A, B or C as specified (see 6.2).

5.1.1 Level A. Unless otherwise specified, the glue shall be packed in 100-pound (45-kg) sacks conforming to 17 to 17X, MB 1 of UU-S-48, or when specified in 50-pound (22.6-kg) sacks conforming to 9 or 9X, MB 1 of UU-S-48 (see 6.7).

5.1.2 Level B. Unless otherwise specified, the glue shall be packed in 100-pound (45-kg) sacks conforming to 15 or 15X, MB. 1 of UU-S-48, or when specified in 50-pound (22.6-kg) sacks conformaing to 5 or 5X, MB 1 of UU-S-48 (see 6.7).

5.1.3 Level C. Unless otherwise specified, the glue shall be packed in commercial 100-pound (45-kg) paper sacks or burlap bags, or when specified in 50-pound (22.6-kg) paper sacks or burlap bags (see 6.7). The glue shall be packed in a manner which shall insure arrival at destination in satisfactory condition, and which shall be acceptable tc the carrier at lowest rates. Bags and packing shall comply with the Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.2 Marking. Marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The animal glue is intended primarily for use in the hardening operation in ball propellant manufacture.

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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) When a first article is required for inspection and approval (see 3.1, 4.5 and 6.3).
(c) Whether suitability for use with an explosive is required (see (3.5).
(d) Level of packing required (see 5.1).
(e) Capacity of containers (see 5.1.1, 5.1.2 or 5.1.3).

6.3 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of 7-104.55 of the Armed Services Procurement Regulation. The first article should be a preproduction sample. The first article should consist of 200 pounds (90kg) of glue. The contracting officer should include specific instructions in all procurement instructions regarding arrangements of examinations, test and approval of the first article.

6.4 Use with explosives. Information pertaining to the use of the glue, procured under this specification, with explosives may be obtained from the Commanding General, US Army Armament Research & Development Command, Chief Materials & Manufacturing Div, Fire Control & Small Caliber Weapon Systems Laboratory (DRDAR-SCM-M).

6.5 Performance Tests. No supplier of any glue has the necessary test equipment to accomplish the performance tests which places all of them on an equally competitive plane. Glue suppliers can hardly be expected to purchase the test equipment since it is inherently process equipment in the ball-propellant manufacturing facilities. There is absolutely no restriction of any kind for any potential suppliers of subject material. If on issuance of invitation for bids or any other form of procurement request, suppliers will contact ball-propellant manufacturers, the performance test will be adequately described.

6.6 General-Purpose Glue. When a specification for a general-purpose glue is required, Federal Specification MMM-A-100 should be used.

6.7 Weight of sacks or bags. The Occupational Safety and Health Act of 1971 will require contractors, sometime in the future, to start packing in 50-pound (22.6-kg) sacks or bags instead of the 100-pound (450kg) sacks or bags now being used.

Custodians:
Army - MR
Navy - AS
Air Force - 99

Preparing activity
Army - MR

Review activities:
Army - AR
Air Force - 84

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5. PROBLEM AREAS

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b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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